TEXORA · LECTURE SHEET

Twist in Yarn

Twist is what turns a bundle of parallel fibres into a yarn: it creates the lateral pressure that lets fibres grip each other. This sheet covers twist direction and level, the twist multiplier that links twist to count, and the untwist and untwist–retwist methods of measurement.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 11 · LEVEL 3 TERM 1

1. What twist does

In a freshly drafted strand the fibres lie parallel and the strand has almost no strength — pull it and the fibres simply slide apart. Inserting **twist** wraps the fibres into helices around the yarn axis. Tension on the yarn then pulls the helices tighter, pressing fibres against each other; the resulting friction lets the strand carry load. Consequences of twist level:

- **Strength rises** with twist up to an optimum, then falls: beyond the optimum the fibres themselves lie at so steep an angle to the axis that their contribution to axial strength drops and they break in sequence (**twist–strength curve**).
- **Extension and resilience** rise with twist; the yarn becomes livelier and more elastic — and more prone to snarling (twist liveliness) if unbalanced.
- **Handle and appearance:** low twist = soft, bulky, dull; high twist = hard, lean, wiry (crepe yarns exploit extreme twist); twist also sets lustre and cover.
- **Abrasion resistance and hairiness** generally improve with moderate twist as surface fibres are bound in (Topic 14).

2. Direction — S and Z

Hold the yarn vertically. If the visible helices slope like the middle stroke of the letter **Z** (rising to the right), the yarn has **Z twist** — the normal direction for singles yarns from ring spinning. If they slope like the middle of **S**, it has **S twist**. Plied yarns are usually twisted opposite to their singles (S ply on Z singles) so the torques balance and the plied yarn lies dead. Folding direction matters for fabric: mismatched twist directions between warp and weft change fabric spirality and surface effects.

3. Level of twist — TPI, turns per unit length, and the multiplier

Twist level = turns per unit length — tpi (turns/inch), turns/cm, or turns/m

The twist needed for a given effect rises as yarn gets finer, because a fine yarn has fewer fibres to grip. Practice therefore specifies twist through the **twist multiplier (twist factor)**, which holds the helix angle approximately constant across counts:

Twist multiplier (English) TM = tpi / √Ne · tex twist factor α = (turns/cm) × √tex

Both forms express the same idea in different count systems; the numerical ranges differ between systems, so a multiplier value is meaningless until its system is named. Convert the twist itself (tpi ↔ turns/m ↔ turns/cm) first, then apply the multiplier of the new system.

Typical English twist multipliers for cotton ring yarns: weft ≈ 3.0–3.5, warp ≈ 3.75–4.5, hosiery ≈ 2.5–3.0, crepe 5 and above. (Figures are starting points; fibre, machinery and end use move them.)

> **WORKED EXAMPLE — TWIST FROM THE MULTIPLIER**
>
> A warp yarn of Ne 30 is ordered at TM 4.0. Required twist = 4.0 × √30 = 4.0 × 5.477 = **21.9 tpi** (= 21.9 × 39.37 ≈ **862 turns/m**).
>
> The same TM applied to Ne 20 gives 4.0 × √20 = **17.9 tpi**: the coarser yarn needs fewer turns per inch for the same fibre helix angle — the entire point of the multiplier.

4. Twist contraction

Twisting shortens the strand: the fibres travel helically, so the yarn delivered is shorter than the drafted strand fed. **Twist contraction (take-up)** — typically a few percent in spun yarns — must be allowed for in production calculations and shows up in twist testing as the small length change between the twisted and untwisted specimen.

5. Measurement of twist

5.1 Untwist (direct counting) method — the reference for singles yarn

1. A specimen of known gauge length (commonly 25 cm / 10 in) is clamped in a **twist tester**: one fixed jaw, one rotating jaw with a revolution counter; a small standard pretension is applied (enough to straighten, not stretch, the yarn).
2. The rotating jaw turns in the untwisting direction while a needle probe checks the fibres; when the fibres lie parallel — the needle passes freely along the specimen — all twist is removed.
3. Turns counted ÷ gauge length = twist per unit length. Direction is read from the sense of rotation (or from the Z/S inspection above).

5.2 Untwist–retwist method — the routine method

Faster, and the basis of automatic twist testers: the specimen is untwisted and then retwisted in the opposite direction until it returns to its original length (detected by a pointer / length sensor). The total revolutions are **twice** the twist in the specimen (untwist turns + equal retwist turns), so:

$$\text{Turns in specimen} = \text{total revolutions counted} / 2$$

The method assumes the retwist equals the untwist at the original length — a sound working assumption for spun singles yarns, less exact for filament and highly lively yarns, for which the direct method governs.

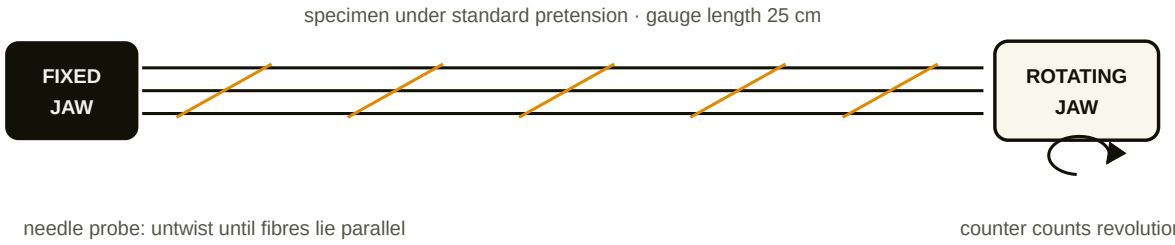


Figure 1 — Twist tester arrangement. Fixed jaw, rotating jaw with counter, defined gauge length and pretension; the probe detects the parallel-fibre end point in the direct method.

WORKED EXAMPLE — UNTWIST–RETWIST CALCULATION

A 25 cm specimen on an untwist–retwist tester records 274 total revolutions to return to its original length.

Turns in the specimen = 274 / 2 = 137 turns in 25 cm = 5.48 turns/cm = **13.9 tpi** (5.48 × 2.54).

If the yarn is Ne 20, its twist multiplier is 13.9 / √20 = 13.9 / 4.472 = **3.11** — a weft-level twist, softer than a warp yarn of the same count should be; if this lot was ordered as warp, the spinning frame's twist change gear is wrong, and the test has paid for itself.

5.3 Twist in plied and filament yarns

For plied yarns the direct method is used with the components separated after measuring ply twist; singles twist is then measured on the unravelled components (it will differ from the original singles twist because plying untwists/twists the singles slightly). Continuous-filament yarns, with low twists, are measured on longer specimens to keep counting error small.

6. Twist variation

- Twist varies along and between yarns through spindle speed variation, traveller behaviour and drafting fluctuations; report mean and CV% from at least 20–30 specimens drawn across packages.
- A subtle trap: twist redistributes itself toward thin places (thin sections accept twist more easily and show *higher* local tpi; thick places show lower). Twist and evenness (Topic 13) are therefore coupled, and hard spots in fabric often trace to twist variation, not count variation.
- Pretension in testing must follow the standard for the yarn type: too much tension extends the specimen and under-reads turns per unit length.

Glossary

Twist — turns about the yarn axis per unit length. **S / Z twist** — helix direction matching the middle stroke of S / Z. **Twist multiplier (TM)** — tpi / √Ne; keeps helix angle constant across counts. **Twist contraction** — shortening of the strand caused by twisting. **Untwist method** — counting revolutions to remove all twist (reference). **Untwist–retwist method** — untwist then retwist to original length; revolutions ÷ 2. **Twist liveliness** — the tendency of an unbalanced twisted yarn to snarl on release.

Key points

• Twist creates strength through fibre-to-fibre friction; strength peaks at an optimum twist, then declines.
• Specify twist by multiplier, not raw tpi, when counts change: tpi = TM × √Ne.
• Reference measurement: direct untwist to parallel fibres; routine: untwist–retwist, total turns ÷ 2.
• Ply opposite to singles for a balanced yarn; report direction, mean and CV% with every twist figure.

Practice

1. Find the tpi for a Ne 16 hosiery yarn at TM 3.0, and for a Ne 16 warp yarn at TM 4.25.
2. An untwist test on a 10 in specimen counts 186 turns to reach parallel fibres. Find the tpi and, for Ne 24, the TM.
3. Why does a plied yarn made Z-on-Z (ply twisted in the same direction as its singles) tend to be lively and unbalanced?
4. Two yarns both show 20 tpi; one is Ne 20, the other Ne 40. Which has the higher helix angle and the harder handle? Show the multipliers.

Answer key

1. Hosiery: 3.0 × √16 = **12.0 tpi**. Warp: 4.25 × 4 = **17.0 tpi**.
2. tpi = 186/10 = **18.6 tpi**. TM = 18.6/√24 = 18.6/4.899 = **3.80**.

3. Plying in the same direction adds torque instead of cancelling it: the ply twist tightens the singles further, storing torsional energy that the finished yarn releases as snarling and skew. Opposite-direction plying lets the singles' residual torque oppose the ply torque, giving a balanced, dead yarn.
4. TM(Ne 20) = 20/√20 = **4.47**; TM(Ne 40) = 20/√40 = **3.16**. The Ne 20 yarn has the higher multiplier, hence the steeper helix angle and harder handle — raw tpi comparisons across counts mislead, which is why the multiplier exists.
